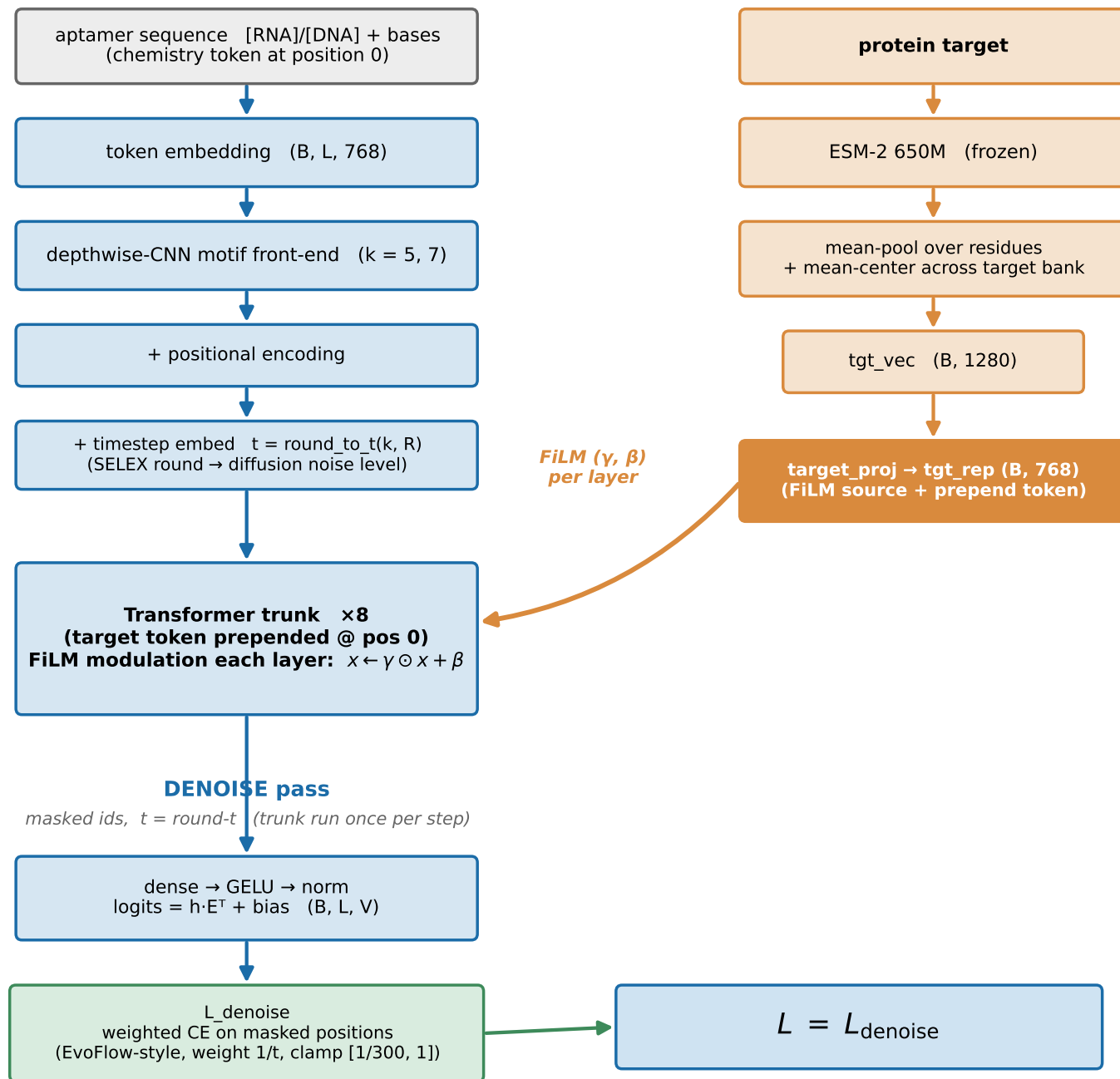


Tumbleweed architecture

75.5M parameters · $d = 768$ · 8 layers · 12 heads · $\text{dim}_{\text{ff}} = 3072$



(contrastive term ablated off; see Table S3)

 generative trunk / head

 frozen ESM-2 target conditioning

 training loss